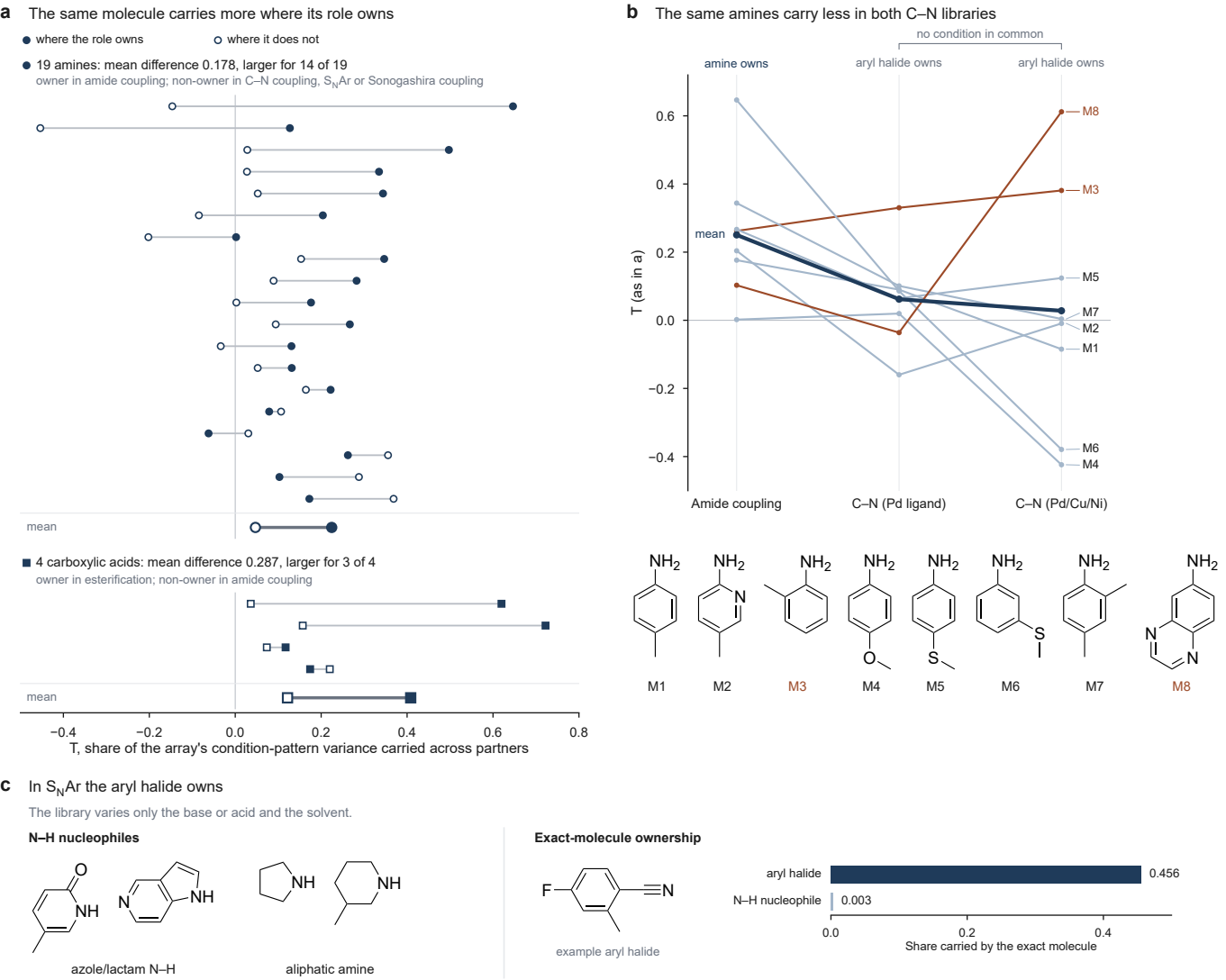

a
The same molecule carries more where its role owns
where the role owns
where it does not
19 amines: mean difference 0.178, larger for 14 of 19
owner in amide coupling; non-owner in C–N coupling, SNAr or Sonogashira coupling
mean
4 carboxylic acids: mean difference 0.287, larger for 3 of 4
owner in esterification; non-owner in amide coupling
mean
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
T, share of the array's condition-pattern variance carried across partners
b
The same amines carry less in both C–N libraries
no condition in common
amine owns
aryl halide owns
aryl halide owns
T (as in a)
0.6
0.4
0.2
0.0
−0.2
−0.4
mean
M8
M3
M5
M7
M2
M1
M6
M4
Amide coupling
C–N (Pd ligand)
C–N (Pd/Cu/Ni)
M1
M2
M3
M4
M5
M6
M7
M8
c
In SNAr the aryl halide owns
The library varies only the base or acid and the solvent.
N–H nucleophiles
azole/lactam N–H
aliphatic amine
Exact-molecule ownership
example aryl halide
aryl halide
0.456
N–H nucleophile
0.003
0.0
0.2
0.4
Share carried by the exact molecule